

The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, white). At the bottom, a Windows taskbar is visible with several application icons including Chrome, Photoshop, and others. The overall image is out of focus, serving as a backdrop for the text.

C#

Level 1

C# Type Casting



C#

Level 1

C# Type Casting

Definition

Type Casting

 Type casting means converting a value from one data type to another.

 C# mainly uses implicit and explicit casting.

Explanation

Implicit Casting

-  Happens automatically when *C#* allows the conversion without requiring a cast.

-  Common example: ``int` → `double``.

Explicit Casting

-  Must be done manually using ``(type)value``.

-  Used when the conversion may lose information or precision.

Procedures

1. Identify the source and target data types.
2. Use implicit casting when *C#* allows the conversion automatically.
3. Use `(targetType)value` for explicit casting.
4. Check the result for possible data loss.

💡 Example

```
int x = 10;  
double y = x;  
double a = 10.75;  
int b = (int)a;
```

➡ Result

Implicit: `10`

Explicit: `10`

Note

- ◆ ``int` → double`` can be done implicitly.
- ◆ ``double` → int`` requires explicit casting.
- ◆ When ``double`` is converted to ``int``, the decimal part is discarded.

⊘ Warning

- ◆ Do not assume that every conversion from a "smaller" type to a "larger" type is automatically safe in terms of precision.
- ◆ Explicit casting does not round decimal values; it removes the fractional part.



Comparison

Implicit Casting

- ✓ Automatic.
- ✓ No cast syntax required.
- ✓ Example: ``double y = x;``

Explicit Casting

- ✓ Manual.
- ✓ Uses ``(type)value``.
- ✓ May cause data loss.
- ✓ Example: ``int b = (int)a;``

✦✦ Summary

- ✓ Implicit casting is performed automatically.
- ✓ Explicit casting uses `(type)value`.
- ✓ Explicit casting can cause data or precision loss.

Thank you 😊